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Early dropout prediction using data mining: a case study with high school students

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Abstract: *Early prediction of school dropout is a serious problem in education, but it is not an easy issue to resolve. On the one hand, there are many factors that can influence student retention. On the other hand, the traditional classification approach used to solve this problem normally has to be implemented at the end of the course to gather maximum information in order to achieve the highest accuracy. In this paper, we propose a methodology and a specific classification algorithm to discover comprehensible prediction models of student dropout as soon as possible. We used data gathered from 419 high schools students in Mexico. We carried out several experiments to predict dropout at different steps of the course, to select the best indicators of dropout and to compare our proposed algorithm versus some classical and imbalanced well-known classification algorithms. Results show that our algorithm was capable of predicting student dropout within the first 4–6 weeks of the course and trustworthy enough to be used in an early warning system.*

Keywords: predicting dropout, classification, educational data mining, grammar-based genetic programming

1. Introduction

Predicting student dropout in high school is an important issue in education because it concerns too many students in individual schools and institutions over the entire world, and it usually results in overall financial loss, lower graduation rates and an inferior school reputation in the eyes of all involved (Neild *et al*., 2007). The definition of dropout differs among researchers, but in any event, if an institution loses a student by whatever means, the institution has a lower retention rate. The early identification of vulnerable students who are prone to drop their courses is crucial for the success of any school retention strategy. And, in order to try to reduce the aforementioned problem, it is necessary to detect students who are at risk as early as possible and thus provide some care in order to prevent these students from quitting their studies and intervene early to facilitate student retention (Heppen & Bowles, 2008). Seidman developed a slogan about student retention (Seidman, 1996) showing that early identification of students at risk, in addition to maintaining intensive and continuous intervention, is the key to reduce dropout levels. So, to develop and use an early warning system (EWS) is a good solution for detecting students at high risk of dropout as early as possible. An EWS is any system that is designed to alert decision makers of potential dangers. Its purpose is to allow for prevention of the problem before it becomes an actual danger (Grasso, 2009). This is a broad definition because there are different types of EWSs that have been used in some areas where detection is important: military attack, conflict prevention, economical/banking crisis, environment disasters/hazards, human and animal epidemics, and so on. In the educational domain, an EWS consists of a set of procedures and instruments for early detection of indicators of students at risk of dropping out and also involves the implementation of appropriate interventions to make them stay in school (Heppen & Bowles, 2008). These indicators are the aspects of students' academic performance that can accurately reflect the risk of dropout corresponding to each of them at a given time. But to detect these indicators or factors is really difficult because there is no single reason why students drop out and in fact, it is a multi-factorial that is also known as the 'one thousand factors problem' (Hernández, 2002). EWSs regularly observe these specific indicators and school performance of students before they drop out. In recent years, effort to create EWSs in education has increased, and nowadays there are some examples of EWSs implemented in different countries:

- The Mexico *Sub-secretary of Middle Education* has defined several guidelines for following the education of young students, and it has developed an EWS based on

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a Microsoft Excel file (Maldonado-Ulloa *et al*., (2011)). This EWS generates alerts starting from three indicators (absenteeism, low performance and problematic behaviour/conduct) with specific critical thresholds that are levels at which it is considered that the probability of dropping out is generally greater.

- The US *National High School Center* has also defined a guide and an EWS (Heppen & Bowles, 2008). It is based on a template from Microsoft Excel and two indicators (course performance and attendance). Starting from this tool, the US *Delaware Department of Education* has implemented an EWS in the states of Chicago, Colorado and Texas (Uekawa *et al*., (2010)). They used a multivariable model to determine which indicators had the strongest correlation with student dropout.
- Finally, in Europe, three countries (Austria, Croatia and England) developed an EWS (Vassiliou, 2013). These systems are focused on systematic monitoring of truancy/absenteeism and results/grades.

After reviewing these EWSs, we think that, on the one hand, to use a simple Excel file (Maldonado-Ulloa *et al*., (2011); Heppen & Bowles, 2008) is not the most appropriate if we have huge amounts of student data available. And on the other hand, statistical techniques have been used for predicting dropout (Uekawa *et al*., (2010); Vassiliou, 2013). Traditionally, statistical models such as logistic regression and discriminant analysis were used most frequently in retention studies to identify factors and their contributions to student dropout (Kovacic, 2010). However, in the last 10 years, Educational Data Mining (EDM) has emerged as a new application area concerned with developing, researching and applying computerized methods to detect patterns in large collections of educational data that would otherwise be hard or impossible to analyse because of the enormous volume of data within which they exist (Baker & Yacef, 2009; Romero & Ventura, 2013). One of the oldest and well-known applications of EDM is predicting student performance in which the goal is to estimate the unknown value of a student's performance, knowledge, score or mark (Romero & Ventura, 2007; Romero & Ventura, 2010; Wolff *et al*., 2014; Yoo & Kim, 2014). Classification is the most commonly employed technique for resolving this problem by discovering predictive models of student performance based on historical data of the students (Hämäläine & Vinni, 2011; Vialardi *et al*., 2011; Romero *et al*., 2013). However, for the early prediction of student dropout, it becomes a harder task because the traditional classification task does not cope well with the temporal nature of this specific kind of data, because it normally considers that all attributes are always available (Antunes, 2010). So, in this paper, we propose a methodology for predicting student dropout as soon as possible, and we also propose an algorithm to obtain a reliable and comprehensible classification model with a sufficiently high accuracy to be used in an EWS. We describe a case study and experiment that we carried out using data from Mexican students in high school education. We want to expose the extent of this problem in Mexico because it is in high school when the dropout rate is the highest of all the educational stages in this country (http://www.snie.sep.gob.mx/) as we can see in Table 1.

The paper is organized as follows: Section 2 shows the most related work about applying data mining for early detection of students at risk of dropout. Section 3 describes our proposed methodology and algorithm. Section 4 presents the data used in the case study, and the experiment carried out is in Section 5. Section 6 shows some examples of models obtained. Section 7 presents a discussion of the results and, finally, Section 8 outlines the conclusions and future work.

2. Background

Tinto's model (Tinto, 1975) is the most widely accepted model in student retention literature. Tinto claims that the decision of students to persist or drop out of their studies is quite strongly related to their degree of academic integration, and social integration, at university. On the other hand, classification algorithms (Kumar & Verma, 2012) are the most widely applied data mining technique for predicting student dropout, as we describe later. A first example of work (Lykourentzou *et al*., 2009) in which several classification techniques [feedforward neural network, support vector machines (SVMs), probabilistic ensemble simplified fuzzy and decision scheme] were applied for dropout prediction in e-learning courses using data from students of the University of Athens. The most successful technique in promptly and accurately predicting dropout-prone students was the decision scheme.

Another comparative analysis of several classification methods (artificial neural networks, decision trees, SVMs and logistic regression) were used in order to develop early models of first year students who are most likely to drop

Table 1: *Dropout rate in Mexico in different educational stages*

Educational stage	*Age of students (years)*	*Dropout 2010–2011 (%)*	*Dropout 2011–2012 (%)*	*Dropout 2012–2013 (%)*
Primary school	6 to 12	0.8	0.7	0.7
Secondary school	12 to 15	5.6	5.4	5.1
High school (preparatoria)	15 to 18	14.5	13.9	13.1
Higher education	Over 18	8.2	8.0	7.9

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out (Delen, 2010). The data for this study come from a public university located in the midwest region of the United States. SVMs performed the best, followed by decision trees, neural networks and logistic regression. Other similar work (Zhang *et al*., 2010) used three classification algorithms (naive Bayes, SVM and decision tree) over university student data in order to improve student retention in higher education. The specific attributes used were the following: average mark, online learning systems information, library information, nationality, university entry certificate, course award, current study level, study mode, age, gender, and so on. Different configurations of the algorithms were tested in order to find the optimum result, and Naive Bayes achieved the highest prediction accuracy, while the Decision Tree had the lowest one. A related work (Kovacic, 2010) used different classification tree methods [Chi-square Automatic Interaction Detector (CHAID), exhaustive CHAID, QUEST and classification and regression tree (CART)] for early prediction of student success. It explored the sociodemographic variables (age, gender, ethnicity, education, work status and disability) and study environment (course programme and course block) that may influence persistence or dropout of students at the *Open Polytechnic* of New Zealand. It found that the most important factors separating successful from unsuccessful students were as follows: ethnicity, course programme and course block; and the most successful classification method was the CART. Class association rules (CAR) were also applied as for predicting student dropout as soon as possible (Antunes, 2010). A CAR is a rule in which the consequent is a single proposition related to the class attribute. The data set used in this study comes from the results of students enrolled in the last 5 years in an undergraduate programme at *Instituto Superior Técnico* in Lisboa. This data set contained 16 attributes about weekly exercises, tests and exams. Finally, several classification models: C4.5 decision tree, naive Bayes, neural networks and rule induction [Repeated Incremental Pruning to Produce Error Reduction algorithm] were used to predict retention in first year and to find the most common factors that influence students in staying or leaving the university (Djulovic & Li, 2013). The conclusion was that it is impossible to say that one model is better than the other one because different performance metrics need to be taken into account. They also found some differences according to some research, in the factors that have more influence in the retention of students, particularly in residency status, gender, age and students' pre-college academic standings.

After reviewing background research (Table 2), we can see that this previous work was applied in higher education (tertiary education) but not in compulsory education (elementary, middle and high school). On the other hand, we can see that there is little consensus on the best method or algorithm to address the dropout problem; while some studies report a particular algorithm as the best performing, for others it is the opposite. The results obtained by these algorithms vary from 65% to 89% accuracy. Traditional classification algorithms are designed to build prediction models based on balanced data sets, that is, there are a similar number of instances/examples/students from some classes than others. However, with regard to the dropout prediction of students, data sets are unbalanced because normally, most of the students continue the course and only a few drop out. In such conditions, accuracy may be misleading because a majority class default classifier would obtain high accuracy, whereas the minority class is mainly ignored. Therefore, it is necessary to design specific algorithms capable of focusing on the minority classes, and in our case of educational data, on the dropout cases, which are what interests us most. The problem of

Table 2: *Background papers*

Work	Subject	Data mining technique	Results
Lykourentzou *et al*., 2009	To predict dropout in e-learning courses	Feedforward neural network, support vector machines, probabilistic ensemble fuzzy and decision scheme	85% Accuracy with decision scheme
Delen, 2010	To predict student retention in university	Artificial neural networks, decision trees, support vector machines and logistic regression	87.23% Accuracy with support vector machines
Zhang *et al*., 2010	To identify potential student at risk in higher education	Naive Bayes, support vector machine and decision tree	89% Accuracy with naive Bayes
Kovacic, 2010	To identify students at risk of dropping out in higher education	CHAID, exhaustive CHAID, QUEST and CART	650.4% Accuracy with CHAID
Antunes, 2010	To anticipate undergraduate student's failure as soon as possible	CAR	80% Accuracy with CAR
Djulovic and Li, 2013	To predict freshman retention in university students	C4.5 decision tree, naive Bayes, neural networks and rule induction	86.27% Accuracy with rule induction

CHAID, Chi-square Automatic Interaction Detector; CART, classification and regression tree; CAR, class association rules.

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imbalanced classes is a challenging task that has received growing attention in recent years (López *et al*., 2013a, He & Garcia, 2009). These methods, based on data resampling and cost-sensitive learning, improved data classification of the minority class while keeping a high overall geometric mean (GM). In particular, much recent research has focused on resampling algorithms and cost-sensitive methods (Galar *et al*., 2012). Data resampling modifies the train data set by adding instances belonging to the minority class to produce a more balanced data class distribution. SMOTE (Chawla *et al*., 2002) is a very well known and commonly employed resampling method that has been shown to improve imbalanced classification, especially when combined with C45 and SVM (López *et al*., 2013b). On the other hand, cost-sensitive learning (López *et al*., 2012) takes into account the misclassification error with respect to the other classes. It thus employs a cost matrix that represents the penalties of misclassifying a given class. Typically, the cost is related to the imbalance ratio (IR) (ratio of sizes of the majority and minority class) to penalize errors happening in the minority class.

3. Proposed methodology and algorithm

The traditional methodology for predicting student dropout uses all the data available at the end of the course together with classical and well-known classification algorithms. Next, we propose both a new methodology and a specific algorithm that attempts to detect students' dropout as early as possible.

3.1. Methodology

Our methodology tries to show at what date or step of the course there is enough data to do a trustworthy enough prediction, good enough to use in an EWS. Different prediction models can be obtained starting from the data gathered at different steps of the course (Figure 1).

As we can see in Figure 1, even at the beginning of the course, an early dropout prediction can be made by using only the data available from previous courses and personal and administrative information about the student. As the course progresses, more information progressively becomes available about the attitudes, activities and performance of students. Therefore, there is no need to wait until the end of the course in order to predict whether a student will continue to the next course or drop out. The actual problem is to determine an early stage in which the prediction is trustworthy enough. The sooner the prediction can be made, the sooner the relevant parties can react and provide specific help to students at risk of dropout in order to try to correct the student's attitude, behaviour and performance in time. In order to detect this step, we propose to apply a specific classification algorithm for obtaining prediction models in each of the steps (Prediction 0 to N-1) using all the available variables/attributes about the students or only the most relevant attributes. We propose to use an Interpretable Classification Rule Mining (ICRM) algorithm instead of traditional classification algorithms. Then, different classification performance measures can be used for determining the earliest step that can be trusted. Finally, at each step, our algorithm obtains accurate and comprehensible classification models based on IF–THEN rules. Starting from the information provided by these discovered models, stakeholders can make decisions concerning students that are predicted to drop out.

3.2. Algorithm

Genetic programming (GP) is an evolutionary algorithm-based methodology used to find computer programs that perform a user-defined task. It is a machine learning technique used to optimize a population of computer programs according to a fitness landscape determined by a program's ability to perform a given computational task. GP has been applied with success in various complex optimization, search and classification problems (Espejo *et al*., 2010; Pan, 2012). The evolutionary algorithm proposed in our work is a variant of GP known as grammar-based genetic programming (GBGP) in which a grammar is defined and the evolutionary process proceeds, guaranteeing that every individual generated is conformant to the grammar (Whigham, 1996). The main advantages of GBGP are its simplicity and flexibility to make the knowledge extracted (rules) more expressive and flexible.

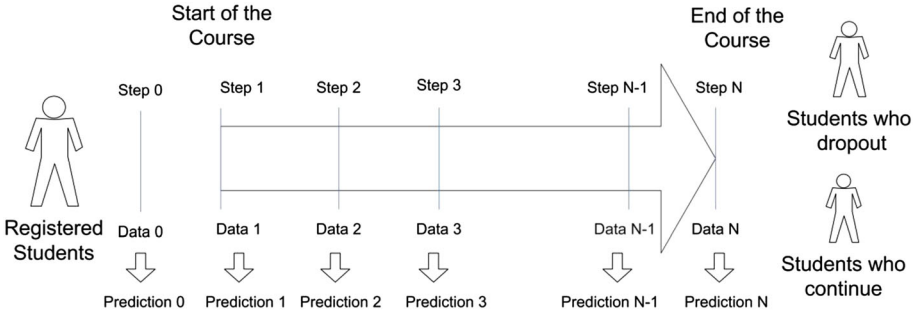


Figure 1: *Proposed dropout prediction methodology.*

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Rules are generated by means of a context-free grammar that defines the production rules, terminal and non-terminal symbols. In this way, classification rules are learned from scratch by appending attribute–value comparisons that improve classification accuracy. More specifically, they improve the value of the fitness function, which is detailed in the following paragraphs.

There are many classification algorithms that provide high levels of accuracy [neural networks, SVM, k-nearest neighbours, etc.], but they are black-box classifiers, that is, it is not feasible to provide the user with the information that leads to predictions. Therefore, the knowledge within the data remains hidden to the expert and the final users. On the one hand, rule-based classifiers provide comprehensible information that shows the knowledge extrapolated from data in the form of understandable IF–THEN classification rules. On the other hand, GP has been employed with success for learning classification rules (Espejo *et al*., 2010). GP is a flexible and powerful evolutionary technique that offers two interesting advantages to classification. The first is its flexibility, which allows the technique to be adapted to the needs of each particular problem. The other is its interpretability because it can employ more interpretable representation formalism, like rules. GBGP is a variation of the classical GP method and, as its name indicates, the main difference among GBGP and GP is that the former uses a grammar to create the population of candidate solutions for the targeted problem. GBGP has been used in a variety of application domains (Pappa & Freitas, 2009) and specifically to the problem of evolving rule sets (Ngan *et al*., 1998; O'Neill *et al*., 2001; Tsakonas *et al*., 2004; Hetland & Saetrom, 2005; Luna *et al*., 2014). With this in mind, we propose a GBGP algorithm for accurate and comprehensible early dropout prediction. This algorithm is a modified version of our previous ICRM algorithm (Cano *et al*., 2013) that we named ICRM2 (see pseudocode in Appendix). Our previous ICRM algorithm already demonstrated to achieve better performance on obtaining accurate and shorter classification rules than other already available algorithms. Hereby, we thought that it can be very useful to use it in the educational data mining context, where the end users are not experts in data mining and they really need comprehensible classification models. Therefore, we adapted it to the early dropout detection problem with unbalanced data. We modified the ICRM algorithm in order to adapt its performance to imbalanced data classes and to focus more specifically on dropout students. Therefore, the new algorithm is mainly focused on obtaining multiple accurate classification rules for predicting which students are going to drop out. The ICRM model is selected as a base model because it showed a very good performance on a wide variety of general-purpose data sets from the University of California, Irvine machine learning repository (Cano *et al*., 2013), achieving high accuracy while providing simple classification rules with a low number of conditions. The latter is very useful for teachers to understand the knowledge in data, as simple classifiers are easily comprehensible. Specifically, the ICRM methodology also demonstrated its advantages when applied to educational data mining problems and in a concrete manner to student failure prediction at school using imbalanced data classification (Marquez-Vera *et al*., 2013). Therefore, owing to its advantages and previous successful application to educational data, we explore its application to early dropout prediction in this paper. The ICRM methodology has been adapted to focus on the prediction of early dropout students where there is less information available about the students. Furthermore, the rule generation procedure of the ICRM2 algorithm is adapted to generate sets of rules focusing on the imbalanced data class (students dropout), which is detailed next. Primarily, our algorithm generates two sets of rules: the former shows the rules that predict student success, whereas the latter predicts student dropout, which most interests us. This is a significant difference between the original ICRM and ICRM2, because the original algorithm obtained only one rule per class, and we are interested in obtaining a full rule set. Generally, only one rule per class is sufficient for accurate classification on general purpose data sets (Cano *et al*., 2013). However, multiple rules allow for obtaining complementary information from squeezing data on several rules covering different sets of attributes. Rules are obtained by means of a GBGP procedure that involves an evolutionary system that uses student data and iteratively constructs classification rules. Evolutionary algorithms codify an individual as a solution to the problem and involve a population of individuals to improve the quality of the solution by means of genetic operators (mutation and crossover). Crossover combines information from two rules to produce a new rule that is expected to improve the previous ones. Mutation introduces new genetic information into the rules (new conditions) so that it provides diversity and advocates exploration of new conditions.

The algorithm iterates to find the best rules that predict student success and dropout using an individual = rule representation, following the genetic iterative rule learning approach. This representation provides greater efficiency and addresses the cooperation–competition problem within the evolutionary process. We use the next context-free grammar to specify which relational operators are allowed to appear in the antecedents of the rules and which attribute must appear in the consequents or class:

$$
\begin{aligned}
&\langle S\rangle \rightarrow \langle comparison\rangle \mid \langle comparison\rangle\ AND\ \langle S\rangle\\
&\langle comparison\rangle \rightarrow \langle operator\rangle\ \langle attribute\rangle\ \langle value\rangle\\
&\langle operator\rangle \rightarrow = \mid \neq\\
&\langle attribute\rangle \rightarrow \textit{any attribute in the data set}\\
&\langle value\rangle \rightarrow \textit{a given value for the attribute}
\end{aligned}
$$

The use of a grammar provides the expressiveness, flexibility and ability to restrict the search space in the search for rules. Rules are generated by the following

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grammar's production rules so that any combination of attribute–value comparisons can be learned and adapted to the data set. Rules are initialized from the initial symbol <S>, and then, they are expanded using the production rules of the grammar, randomly transforming non-terminal symbols into terminal symbols. In this way, a population of diverse rules representing a variety of conditions can be easily created as the algorithm's initial population. The genetic operators are then applied to improve and combine the rules' conditions and evaluated according to the fitness function. The implementation of constraints using a grammar can be a very natural way to express the syntax of rules when individual representation is specified. The relational operators for nominal attributes are 'equal' (=) and 'not equal' (≠). These rules can be applied to a great number of learning problems. The rules are constructed to find the conjunction of conditions on the relevant attributes that best discriminates a class from the other classes. The key for learning good rules for a given problem is defining a proper fitness function.

The fitness function (equation (1)) evaluates the quality of the represented solutions for maximizing the classification performance regardless of whether or not the data are imbalanced. The definition of a proper fitness function is crucial for imbalanced data classification using GP (Patterson & Zhang, 2007). The function searches for rules that maximize both sensitivity and specificity simultaneously, evaluating complementary aspects of the positive/negative class errors. This can be carried out by multiplying the two independent measures to acquire a single-valued measure that guides the evolutionary process. In our case, it lets us find rules that truly predict student dropout while not producing a high number of prediction errors and not missing other students that are likely to fail.

$$\text{Fitness} = \text{Sensitivity} * \text{Specificity} \quad (1)$$

We used a combination of two measures that are commonplace in classification. On the one hand, specificity (equation (2)) focuses on improving the performance of the dropout prediction, measuring the number of truly detected dropout cases and the missing cases. And on the other hand, sensitivity (equation (3)) balances the number of truly predicted success cases and the number of false negative dropout cases.

$$\text{Specificity} = TN/TN + FP \quad (2)$$

$$\text{Sensitivity} = TP/TP + FN \quad (3)$$

These measures are calculated by means of confusion matrix values (Table 3).

Finally, by means of using this fitness function (equation (1)), our algorithm is aimed to search for rules that maximize both sensitivity and specificity. In our case, it finds rules that truly predict student dropout while not producing a high number of prediction errors and not missing other students that are likely to drop out. Therefore, it seeks a balance between the classes and a trade-off for predicting both classes correctly, taking into account that if classes are imbalanced, the positive/negative ratios will indicate this behaviour so that the evolutionary process will lead to rules with better trade-offs.

Table 3: *Confusion matrix*

Actual versus predicted	*Positive*	*Negative*
Positive	TP	FN
Negative	FP	TN

TP, true positive; TN, true negative; FP, false positive; FN, false negative.

4. Data set

The data set used in this work comes from 419 students enrolled in the Academic Unit Preparatoria at the Autonomous University of Zacatecas in Mexico. All students were about 15 years old and were registered in first year of the preparatoria (high school). In this study, we used only the information of the first semester, that is, when more students drop out. In fact, in our case, 13.6% of students drop out, as we can see in Figure 2.

All the data used have been gathered from different sources and on different occasions during the period from August to December 2012. Figure 3 shows the specific steps when the student information was gathered. We used these stages for collecting the information in accordance with the particular characteristics of Mexico Academic Program II. But our proposed methodology can be implemented in other institutions by simply changing the number of stages and dates depending on their own characteristics.

Step 0 was before the beginning of the semester, and it contained previous marks/scores. At this stage, the only available information about students came from the admission exam. Step I was just at the beginning of the semester and had general information about school enrolment. Once students were enrolled, we obtained new

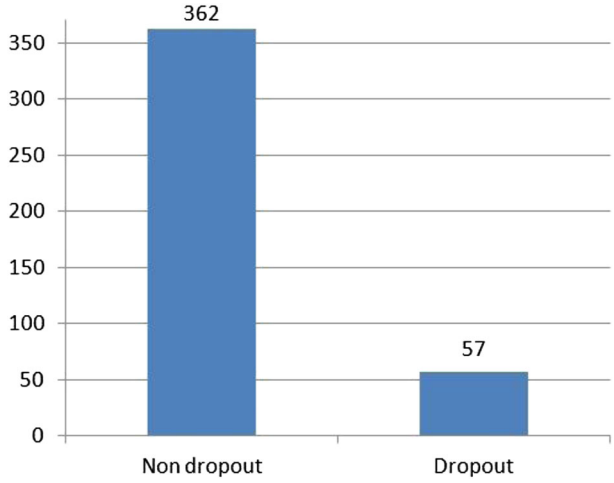


Figure 2: *Distribution of student dropout.*

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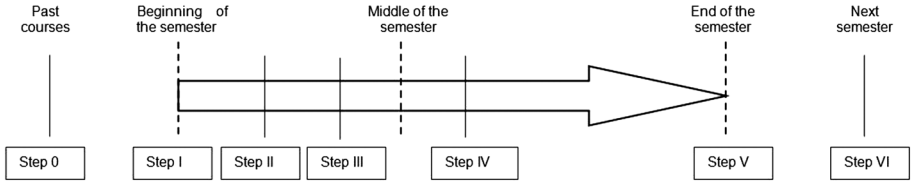


Figure 3: *Steps in which data are gathered.*

information from their registration. Step II was 4 weeks after the beginning of the semester, and it had information about some conditional physical abilities. An evaluation of student's physical abilities was carried out by physical education teachers. Step III was 6 weeks after the beginning of the semester, and it had information about attendance and student behaviour. Teachers of each group provided this information about students who attended their class. Step IV was 10 weeks after the beginning of the semester, and it had a great amount of information about other factors that could affect school performance. This information was collected by means of a survey (Marquez-Vera *et al*., 2013) distributed to all students. Step V was at the end of the semester and outlined the final scores obtained by students in all subjects. Teachers reported the student's final grades to the school. And finally, Step VI was just before the beginning of the next semester, and it provided information about which students enrol in the next semester and which students drop out. The specific information or attributes used in each step is shown in Table 4.

As we can see in Table 4, there are a total of 60 attributes or indicators available gathered in different steps (from 0 to V) in order to predict which students drop out or continue to the next semester (Step VI).

5. Experiments

We carried out three experiments in order to test our methodology and to compare the performance of our proposed ICRM2 algorithm versus five classical and four imbalanced well-known classification algorithms publicly available in WEKA data mining software (Witten *et al*., 2011).

5.1. Experiment 1

In this first experiment, we predicted dropout by using all the attributes in each step of the course, that is, all attributes available from the beginning of the course in the corresponding stages. We executed the following classical classification algorithms:

- *Bayesian classifier*, NaiveBayes (John & Langley, 1995). A naive Bayes classifier is a simple probabilistic classifier based on Bayes' theorem with strong (naive) feature

Table 4: *Student information used in each step*

Step	*N*. attributes	Name/description of attributes added in each step
0	2	Grade point average in secondary school and average score in EXANI I
I	10	Classroom/group enrolled, size of the classroom, age, attendance during morning/evening sessions, family income level, having scholarship, having a job, living with one's parents, mother's level of education and father's level of education
II	11	Having a physical disability, height, weight, waist, measure of flexibility, abdominal exercises in a minute, push-ups in a minute, time in 50-m race, time in 1000-m race, regular consumption of alcohol and smoking habits
III	4	Attendance, level of boredom during classes, misbehaviour and having an administrative sanction
IV	26	Number of friends, number of hours spent studying daily, group studying, place normally used for studying, study habits, way to resolve doubts, level of motivation, religion, external influence in choice of degree, personality type, resources for studying, number of brothers/sisters, position as the oldest/middle/youngest child, parental encouragement for study, number of years living in a city, transport method used to go to school, distance to school, interest in the subjects, level of difficulty of the subjects, taking notes in class, too heavy a demand of homework, methods of teaching, quality of school infrastructure, having a personal tutor and level of teacher's concern for the welfare of each student
V	7	Score in Maths, score in Physics, score in Social Science, score in Humanities, score in Writing and Reading, score in English and score in Computer Science
VI	1	Who drop out/continue in the next semester

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independence assumptions. In simple terms, a naive Bayes classifier assumes that the presence or absence of a particular feature is unrelated to the presence or absence of any other feature, given the class variable.

- S*VM*, sequential minimal optimization (SMO) (Platt, 1998) implements Platt's SMO algorithm for training a support vector classifier using polynomial or radial basis function kernels. This implementation globally replaces all missing values and transforms nominal attributes into binary ones. It also normalizes all attributes by default. Multi-class problems are solved using pairwise classifiers.
- *Instance-based lazy learning*, IBk (Aha & Kibler, 1991). The well-known KNN algorithm classifies an instance with the class with the highest value of the number of neighbours to the instance that belongs to such class.
- *Classification rules*, JRip (Cohen, 1995). Implements a propositional rule learner, Repeated Incremental Pruning to Produce Error Reduction, which was proposed by William W. Cohen as an optimized version of Incremental Reduced Error Pruning. It is based on association rules with reduced error pruning, a very common and effective technique found in decision tree algorithms.
- *Decision trees*, J48 (Quinlan, 1993). J48 is the open source implementation of the C4.5 algorithm. C4.5 builds decision trees from a set of training data using the concept of information entropy. At each node of the tree, C4.5 chooses the attribute of the data that most effectively splits its set of samples into subsets enriched in one class or the other. The attribute with the highest information gain is chosen to make the decision. The C4.5 algorithm then resorts to the smaller subset.

To evaluate the performance of the classifiers at each step of the course, the next well-known measures (provided by the confusion matrix) are used:

- *Accuracy* (*Acc*) is the overall accuracy rate or classification accuracy and is calculated as follows:

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \tag{4}$$

- *True positive rate* (*TP rate*) or sensitivity or recall is the proportion of actual positives that are predicted positive. We use the TP rate to measure the successful students, and it is calculated as follows:

$$TP\,\text{rate} = \frac{TP}{TP \;+\; FN} \tag{5}$$

- *True negative rate* (*TN rate*) or specificity is the proportion of actual negatives that are predicted negative. We use the TN rate to measure the dropout students, and it is calculated as follows:

$$TN\ \text{rate} = \frac{TN}{TN \;+\; FP} \tag{6}$$

- *GM* indicates the balance between two classification measures. It represents a trade-off measure commonly used with imbalanced data sets and is calculated as follows:

$$GM = \sqrt{TP_{\text{rate}} \cdot TN_{\text{rate}}} \tag{7}$$

We executed all classification algorithms using a 10-fold cross-validation procedure in which all executions are repeated 10 times using different train/test partitions of the data set using the WEKA's procedure for cross-validation (Witten *et al.*, 2011). The 10-fold cross-validation procedure divides the data set into 10 roughly equal parts. For each part, it trains the model using the nine remaining parts and computes the test error by classifying the given part. Finally, the results for the 10 test partitions are averaged. These test classification results obtained with all the algorithms are shown in Table 5.

In the beginning (step 0), only two attributes were known. This information was gathered before the beginning of the course, and it indicates the performance of the students in previous courses and exams. The dropout prediction (TN)

Table 5: *Classification results in each step using all the attributes*

	Step 0	Step I	Step II	Step III	Step IV	Step V
TP rate						
NaiveBayes	0.994	0.873	0.931	0.909	0.901	0.961
SMO	0.992	0.981	0.961	0.931	0.928	0.992
IBk	0.994	0.948	0.961	0.981	0.967	0.986
JRip	0.994	0.983	0.950	0.950	0.959	0.981
J48	1.000	1.000	0.975	0.967	0.981	0.983
ICRM	0.735	0.769	0.876	0.975	0.981	1.000
TN rate						
NaiveBayes	0.070	0.298	0.509	0.719	0.649	0.965
SMO	0.000	0.018	0.544	0.632	0.561	0.912
IBk	0.070	0.123	0.579	0.561	0.421	0.895
JRip	0.070	0.000	0.439	0.614	0.649	0.842
J48	0.000	0.000	0.474	0.579	0.544	0.807
ICRM	0.807	0.825	0.843	0.857	0.895	0.983
Accuracy						
NaiveBayes	0.869	0.795	0.874	0.883	0.866	0.962
SMO	0.857	0.850	0.905	0.890	0.878	0.981
IBk	0.869	0.835	0.909	0.924	0.893	0.974
JRip	0.869	0.850	0.881	0.905	0.916	0.962
J48	0.864	0.864	0.907	0.914	0.921	0.959
ICRM	0.733	0.782	0.857	0.945	0.950	0.998
GM						
NaiveBayes	0.264	0.510	0.688	0.808	0.765	0.963
SMO	0.000	0.133	0.723	0.767	0.722	0.951
IBk	0.264	0.341	0.746	0.742	0.638	0.939
JRip	0.264	0.000	0.646	0.764	0.789	0.909
J48	0.000	0.000	0.680	0.748	0.731	0.891
ICRM	0.770	0.797	0.859	0.914	0.937	0.991

TP, true positive; SMO, sequential minimal optimization; ICRM, Interpretable Classification Rule Mining; TN, true negative; GM, geometric mean.

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and GM values obtained by the ICRM2 algorithm (Table 5) were the highest with much difference with the other algorithms. However, the ICRM2 algorithm obtained a low value in the general accuracy and in predicting students who actually continue in the next semester (TPs). Because of this fact, this step and algorithm is not recommended for early prediction. On the other hand, the rest of the algorithms achieved a very high TP ratio (very close to 1.0) but with the high cost of having a very low dropout prediction (lower than 0.1), that is, they predicted directly almost all students as set to continue. Therefore, the predictions of the classical algorithms at this step should not be trusted because of its high inaccuracy between classes.

In Step I, 10 more attributes were gathered and added to the data set providing more information about the class statistics, attendance and social information about students. The new information allowed for increasing the TP ratio a little for the ICRM2 algorithm while keeping a high TN prediction. However, this TP value was not high enough to be used in a trusted early prediction system. In the opposite site, some of the other algorithms increased their TN ratio only a little but decreased their TP ratio.

In Step II, 11 more attributes with information about the physical conditions of students were added to the data set. This time, the ICRM2 significantly reduced the distance with the other algorithms with regard to the TP ratio (higher than 0.95) but maintained the highest dropout prediction. And the rest of the algorithms significantly increased the dropout prediction (close to 0.5) and, consequently, the GM is improved to more acceptable levels (close to 0.7). This is the first step in which the performance classification measures are trustworthy enough to make an early prediction of dropout, especially using our ICRM2 algorithm.

In Step III, four attributes about student behaviour in class were appended to the data set. As seen in Table 5, the TN and GM values increased in all the algorithms. In fact, the ICRM2 algorithm obtained a very high TP ratio (higher than 0.95) while maintaining a very high dropout (higher than 0.8). This good performance led us to strongly recommend the use of this algorithm for early prediction of dropout in this step. It is especially noteworthy to mention that this step is before the middle of the course, when there is still time to try to help these students and prevent them from dropping out.

In Step IV, 26 new attributes of information about other factors that could affect school performance were added to the data set. However, as seen in Table 5, these new attributes introduced too much information and noise to all the algorithms' performance. Therefore, most of the algorithms were overwhelmed and their performance decreased, especially with regard to the dropout prediction. Moreover, this step happened after the middle of the course, when it can be a little too late for early prediction.

And the last Step V provided information about the scores obtained in the various exams of the seven subjects of the course. All algorithms were capable of predicting dropout successfully with a high value (near 1). It shows that predicting a student's final status by using exam scores is very obvious and naïve because they are highly correlated. However, this step happened at the end of the course when there is no possibility of any intervention to help students at risk of dropout and therefore, it cannot be used as an early prediction.

Finally, we compared the computational cost of running all the algorithms. The five classical algorithms were executed in less than 1 s as we can see in Table 6. ICRM2 recorded a significant greater time in all steps because of it being an evolutionary-based method. However, as shown in Table 5, its runtime was not prohibitive given the time frame of our problem: in the worse case, it took 19 s to run when using all the attributes in Step V. And its runtime also increased significantly as the number of attributes grew along the steps.

5.2. *Experiment 2*

In the second experiment, we carried out a study of feature selection in order to identify which attributes have a greater effect on our class prediction (dropout or continue) at each step. Our aim is to try and solve the problem of high dimensional data by reducing the number of used attributes without losing reliability in classification. In order to select the best attributes in each step, we repeat for each step the same procedure described in our previous work (Marquez-Vera *et al*., 2013) in which we used 10 attribute selection algorithms provided by WEKA (Witten *et al*., 2011):

- Three attribute subset evaluators (CfsSubsetEval, ConsistencySubsetEval and FilteredAttributeEval) were used for searching the space of attributes subsets, evaluating each one. We used the default search method (BestFirst) for crossing the attribute space to find a good subset.
- Seven single-attribute evaluators (ChiSquaredAttributeEval, OneRAttributeEval, FilteredSubsetEval, GainRatioAttributeEval, InfoGainAttributeEval, ReliefFAttributeEval and SymmetricalUncertAttributeEval) were used for

Table 6: *Execution time (in seconds) when using all attributes*

Algorithm	Step 0	Step I	Step II	Step III	Step IV	Step V
NaiveBayes	0.01	0.01	0.01	0.02	0.02	0.03
SMO	0.08	0.09	0.17	0.20	0.25	0.28
IBk	0.01	0.01	0.01	0.01	0.01	0.02
JRip	0.01	0.06	0.06	0.08	0.08	0.11
J48	0.02	0.02	0.06	0.06	0.06	0.08
ICRM2	0.11	1.11	5.92	8.52	13.23	19.02

SMO, sequential minimal optimization; ICRM, Interpretable Classification Rule Mining.

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evaluating the attributes individually and sorting them. We used the only provided ranking method (Ranker) to rank individual attributes (not subsets) according to their evaluation.

At each step, we executed the 10 feature selection algorithms using only the attributes of the corresponding step.

On the one hand, the three attribute subset evaluators returned a list of selected attributes or subset that is most likely to predict the class best. On the other hand, the seven single-attribute evaluators returned a ranked list of all the attributes. We therefore had to remove the lower-ranking ones in order to perform attribute selection by discarding attributes that fall below a chosen cut-off point. We used as a cut-off point the mean value or average of all the scores of each ranked list of attributes. In this way, the 10 feature selection algorithms returned a subset or list of selected attributes. Finally, in order to obtain the best attributes at each step, we ranked the results obtained by the previous 10 algorithms using the following method: (1) we counted the number of times each attribute was selected by one of the algorithms; and (2) we selected as the best attributes of each step only those with a frequency greater than two, that is, to say, attributes that have been considered by at least two feature selection algorithms. Table 7 shows the list of selected best attributes in each step of the course (the number of attributes, their names and their frequency between brackets).

We can see when comparing the attributes of Table 7 with Table 4 that there is a high reduction of attributes in some steps such as Step II (from 11 to 3) and still more pronounced in Step IV (from 26 to 2).

Then, we executed all the classification algorithms in the same way as in first experiment but using only these selected attributes, that is, all the selected attributes from the beginning of the course until the corresponding step. The test classification results obtained in the second experiment are shown in Table 8.

In the beginning, only the grade point average (GPA) in secondary school was selected as relevant. When comparing results using the best attributes (Step 0 in Table 8) and all the attributes (Step 0 in Table 5), it can be seen that the TP and accuracy values are similar in all the algorithms, whereas the TN and GM values are lower for almost all algorithms but ICRM2.

In Step I, only six attributes (about class properties and social conditions of students) were added to the data set as the most relevant. The results obtained with the four measures of classification performance were similar to those obtained when using all attributes, and again, the ICRM2 algorithm achieved the best results.

In Step II, only three attributes (physical resistance, smoking and alcohol drinking habits) were considered relevant for classification. As seen in Table 8, the increase of the dropout prediction in all algorithms is improved when appending these three new attributes to the data set, and the increase of TP prediction is especially noticeable in ICRM2 algorithm. So, we can recommend the use of ICRM2 algorithm in this step for making an early prediction of dropout with a very good performance.

In Step III, only two attributes (about class attendance and behaviour sanctions) were considered relevant. All algorithms increased their measures a little, specially the TP and accuracy of the ICRM2 algorithm. So, although this step and this algorithm are strongly recommended for

Table 7: *Best attributes in each step selected by the feature selection algorithms*

Step	N. of at.	Name of attributes added in each step
0	1	Grade point average in secondary school (6)
I	6	Classroom/group enrolled (5), number of students in the group/class (3), age (5), attendance during morning/evening sessions (5), having a job (4) and mother's level of education (2)
II	3	Time in 1000-m race (3), regular consumption of alcohol (6) and smoking habits (4)
III	2	Attendance (5) and having administrative sanction (5)
IV	2	Place normally used for studying (2) and level of motivation (6)
V	3	Score in Maths (6), score in Social Science (5) and score in Humanities (3)

Table 8: *Classification results in each step using the best attributes*

	Step 0	Step I	Step II	Step III	Step IV	Step V
TP rate						
NaiveBayes	1.000	0.854	0.901	0.912	0.925	0.967
SMO	1.000	1.000	0.972	0.972	0.972	0.983
IBk	1.000	0.956	0.972	0.972	0.967	0.978
JRip	1.000	0.989	0.964	0.964	0.953	0.970
J48	1.000	1.000	0.975	0.970	0.978	0.983
ICRM	0.710	0.761	0.925	0.959	0.975	0.978
TN rate						
NaiveBayes	0.000	0.333	0.491	0.614	0.649	0.947
SMO	0.000	0.000	0.439	0.491	0.561	0.842
IBk	0.000	0.123	0.316	0.351	0.421	0.772
JRip	0.000	0.000	0.421	0.596	0.649	0.789
J48	0.000	0.000	0.474	0.561	0.579	0.842
ICRM	0.772	0.789	0.825	0.825	0.842	0.965
Accuracy						
NaiveBayes	0.860	0.783	0.845	0.871	0.888	0.964
SMO	0.860	0.864	0.900	0.907	0.916	0.964
IBk	0.860	0.842	0.883	0.888	0.893	0.950
JRip	0.860	0.854	0.890	0.914	0.912	0.945
J48	0.860	0.864	0.907	0.914	0.924	0.964
ICRM	0.743	0.782	0.900	0.950	0.964	0.976
GM						
NaiveBayes	0.000	0.533	0.665	0.748	0.775	0.957
SMO	0.000	0.000	0.653	0.691	0.738	0.910
IBk	0.000	0.343	0.554	0.584	0.638	0.869
JRip	0.000	0.000	0.637	0.758	0.786	0.875
J48	0.000	0.000	0.680	0.738	0.753	0.910
ICRM	0.740	0.775	0.874	0.889	0.906	0.971

TP, true positive; SMO, sequential minimal optimization; ICRM, Interpretable Classification Rule Mining; TN, true negative; GM, geometric mean.

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making an early prediction, the previous step can be preferable because it obtained a similar or slightly lower performance but in an earlier step.

In Step IV, only two attributes (the location where students use to study and the expectation/self-confidence to pass the course) were selected. It is interesting to note that these two attributes did not decrease the accuracy as was previously observed when adding all the attributes in this step. On the contrary, now they are helpful to increase the dropout prediction in all algorithms. Nevertheless, this step happens later than halfway through the course, and therefore, it may be too late for an early prediction.

In the last step, only three attributes (scores obtained in Maths, Social Science and Humanities) were selected. And, as in the first experiments, almost all algorithms were capable of predicting dropout successfully with a high performance (near 1).

The comparative analysis of the computational cost of the algorithms is particularly interesting when using the selected subset of best attributes (Table 9). It is interesting to note the reduction of the computation time of the ICRM2 algorithm as compared with the previous runtimes showed in Table 6. The smaller number of attributes also allowed for a significant speed-up, which reduced the execution time at Step V to only 3 s, rendering this approach more meaningful.

5.3. Experiment 3

In the third experiment, we compared the performance of our proposed ICRM2 algorithm with four classification algorithms specifically designed for imbalanced data (in our case, there are many more 'continue' than 'dropout' students). These algorithms are based on data resampling and cost-sensitive learning (López *et al*., 2013a):

- *C45-SMOTE*, Data are resampled using SMOTE (Chawla *et al*., 2002) and are then classified by the C45 classifier (López *et al*., 2013b).
- *SVM-SMOTE*, Data are resampled using SMOTE (Chawla *et al*., 2002) and are then classified by the SVM classifier (López *et al*., 2013b).
- *C45-CS*, The cost-sensitive classifier takes into account the cost matrix to build a C45 decision tree (López *et al*., 2012). The used cost matrix is [[0,1],[6,0]]. In other words, there is a significant penalty for misclassifying a minority class example. This value is obtained by measuring the IR of the two data classes, which is about 6. IR is defined by the size of the majority class divided by the size of the minority class.
- *SVM-CS*, The cost-sensitive classifier takes into account the cost matrix [[0,1],[6,0]] to build an SVM classifier (López *et al*., 2012).
- *GP-COACH-H*. Data are resampled using SMOTE (Chawla *et al*., 2002) and is then classified by a hierarchical genetic fuzzy system based on GP (López *et al*., 2013b).

Table 9: *Execution time (in seconds) when using the best attributes*

Algorithm	Step 0	Step I	Step II	Step III	Step IV	Step V
NaiveBayes	0.01	0.01	0.01	0.01	0.02	0.02
SMO	0.03	0.04	0.13	0.14	0.14	0.19
IBk	0.01	0.01	0.01	0.01	0.01	0.01
JRip	0.01	0.04	0.04	0.04	0.05	0.06
J48	0.02	0.02	0.03	0.03	0.05	0.06
ICRM2	0.05	0.20	0.58	0.87	2.02	3.03

SMO, sequential minimal optimization; ICRM, Interpretable Classification Rule Mining.

For a performance evaluation of these classifiers at each step of the course within the context of imbalanced data sets, accuracy is no longer a proper measure, because it does not distinguish between the numbers of correctly classified examples of different classes. A default hypothesis classifier could, in fact, achieve very high accuracy by only predicting the majority class. For example, if a classification model predicts all students to the class of 'continue', the accuracy of the data set is expected to be 86.4% (13.6% of students drop out), which is the statistical distribution of the data. In order to avoid this problem, other different performance metrics such as the GM and AUC (area under the receiver operating characteristic curve) are normally used when dealing with imbalanced data (Fernández *et al*., 2008, Raeder *et al*., 2012). AUC shows the trade-off between the TP rate and the FP rate, and it is calculated as (López *et al*., 2013a)

$$\mathrm{AUC} = \frac{1 + TP - FP}{2} \tag{8}$$

Table 10 shows the GM and the AUC at the different stages when using all attributes. To be noted is the high increase of the GM in the early stages for both the rebalanced and cost-sensitive approaches as compared with the results previously shown in Table 5 without considering the imbalance scenario. Moreover, it is also interesting to highlight that the cost-sensitive approach produces better results than resampling at early stages. However, this behaviour is swapped as more information becomes available in further stages. Thus, at Steps 0 and I, both cost-sensitive methods perform better than their resampling relatives. However, from Step II, the resampling methods show better performance as compared with the cost-sensitive methods. On the other hand, ICRM2 shows a better GM and AUC for all the stages.

Table 11 shows the GM and the AUC for algorithms for imbalanced data when using the selected best attributes. The difference between resampling and cost-sensitive methods is increased in this experiment at Step 0. This is primarily due to the lower number of attributes at the early stage that have been selected. However, as more data become available at Steps III, IV and V, the performance difference between the two cost-sensitive and resampling decreases. On the

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Table 10: *Classification results for imbalanced algorithms using all attributes*

	Step 0	Step I	Step II	Step III	Step IV	Step V
TP rate						
C45-SMOTE	0.983	0.890	0.937	0.937	0.953	0.981
SVM-SMOTE	0.972	0.997	0.997	0.992	0.997	0.995
C45-CS	0.649	0.776	0.854	0.865	0.870	0.959
SVM-CS	0.644	0.782	0.934	0.934	0.950	0.989
GP-COACH-H	0.466	0.748	0.785	0.909	0.922	0.986
ICRM	0.807	0.825	0.843	0.857	0.895	0.983
TN rate						
C45-SMOTE	0.079	0.456	0.623	0.667	0.702	0.833
SVM-SMOTE	0.158	0.173	0.377	0.544	0.483	0.974
C45-CS	0.737	0.474	0.544	0.702	0.702	0.877
SVM-CS	0.737	0.404	0.509	0.702	0.561	0.947
GP-COACH-H	0.666	0.594	0.609	0.842	0.852	0.986
ICRM	0.735	0.769	0.876	0.975	0.981	1.000
Accuracy						
C45-SMOTE	0.767	0.786	0.861	0.872	0.893	0.945
SVM-SMOTE	0.777	0.758	0.849	0.885	0.874	0.990
C45-CS	0.661	0.735	0.812	0.843	0.847	0.948
SVM-CS	0.656	0.730	0.876	0.902	0.897	0.983
GP-COACH-H	0.492	0.728	0.762	0.900	0.913	0.986
ICRM	0.733	0.782	0.857	0.945	0.950	0.998
GM						
C45-SMOTE	0.279	0.637	0.764	0.790	0.818	0.904
SVM-SMOTE	0.392	0.415	0.613	0.734	0.694	0.984
C45-CS	0.692	0.606	0.681	0.779	0.781	0.917
SVM-CS	0.689	0.562	0.689	0.809	0.730	0.968
GP-COACH-H	0.557	0.666	0.691	0.875	0.886	0.986
ICRM	0.770	0.797	0.859	0.914	0.937	0.991
AUC						
C45-SMOTE	0.643	0.769	0.841	0.836	0.873	0.946
SVM-SMOTE	0.565	0.499	0.687	0.768	0.740	0.984
C45-CS	0.702	0.626	0.727	0.790	0.797	0.928
SVM-CS	0.690	0.593	0.721	0.818	0.756	0.968
GP-COACH-H	0.567	0.621	0.749	0.875	0.854	0.984
ICRM	0.787	0.806	0.854	0.923	0.946	0.991

TP, true positive; SMO, sequential minimal optimization; ICRM, Interpretable Classification Rule Mining; TN, true negative; GM, geometric mean; AUC, area under the receiver operating characteristic curve.

Table 11: *Classification results for imbalanced algorithms using best attributes*

	Step 0	Step I	Step II	Step III	Step IV	Step V
TP rate						
C45-SMOTE	1.000	0.950	0.961	0.961	0.964	0.989
SVM-SMOTE	1.000	1.000	0.981	0.992	0.989	0.995
C45-CS	0.613	0.746	0.887	0.928	0.920	0.967
SVM-CS	0.613	0.751	0.939	0.931	0.964	0.981
GP-COACH-H	0.417	0.688	0.895	0.910	0.935	0.986
ICRM	0.772	0.789	0.825	0.825	0.842	0.965
TN rate						
C45-SMOTE	0.000	0.211	0.500	0.649	0.667	0.930
SVM-SMOTE	0.000	0.000	0.412	0.474	0.518	0.939
C45-CS	0.825	0.474	0.439	0.649	0.632	0.895
SVM-CS	0.825	0.456	0.456	0.561	0.632	0.930
GP-COACH-H	0.614	0.607	0.624	0.757	0.891	0.951
ICRM	0.710	0.761	0.925	0.959	0.975	0.978
Accuracy						
C45-SMOTE	0.761	0.773	0.851	0.887	0.893	0.975
SVM-SMOTE	0.761	0.761	0.845	0.868	0.876	0.981
C45-CS	0.642	0.709	0.826	0.890	0.881	0.957
SVM-CS	0.642	0.711	0.874	0.881	0.919	0.974
GP-COACH-H	0.443	0.678	0.860	0.890	0.929	0.981
ICRM	0.743	0.782	0.900	0.950	0.964	0.976
GM						
C45-SMOTE	0.000	0.447	0.693	0.790	0.802	0.959
SVM-SMOTE	0.000	0.000	0.636	0.685	0.715	0.966
C45-CS	0.711	0.594	0.624	0.776	0.762	0.930
SVM-CS	0.711	0.585	0.654	0.723	0.780	0.955
GP-COACH-H	0.506	0.646	0.747	0.830	0.913	0.968
ICRM	0.740	0.775	0.874	0.889	0.906	0.971
AUC						
C45-SMOTE	0.500	0.676	0.830	0.866	0.866	0.976
SVM-SMOTE	0.500	0.500	0.697	0.733	0.753	0.967
C45-CS	0.715	0.555	0.744	0.776	0.771	0.940
SVM-CS	0.719	0.604	0.698	0.746	0.798	0.955
GP-COACH-H	0.516	0.648	0.759	0.846	0.883	0.963
ICRM	0.745	0.771	0.875	0.890	0.906	y84

TP, true positive; SMO, sequential minimal optimization; ICRM, Interpretable Classification Rule Mining; TN, true negative; GM, geometric mean; AUC, area under the receiver operating characteristic curve.

other hand, the ICRM2 algorithm keeps the best GM and AUC results, even with the smaller set of best attributes.

Insofar as computing times are concerned, they are very similar to those of previous experiments because the resampling and cost-sensitive approaches have very small impact on the runtime. This is due to the relatively small size of the data, as SMOTE takes very few milliseconds to create examples for the minority class. In order to avoid text overloading and excessive repetition of similar results, they have been omitted for the third experiment. On the other hand, GP-COACH-H takes several hours, especially as the number of attributes increase in later steps considering more information.

6. Discovered models

Two examples of the different models discovered by our ICRM2 algorithm in each experiment are shown and

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described in the following. The objective is to see their accuracy and usefulness for providing information about students at risk of dropout. Specifically, we show the models discovered at Step II using the best attributes and at the last step using all attributes. This will allow us to compare the rules obtained at an early prediction stage versus the rules obtained at the traditional approach of using all the available information at the end of the course.

6.1. Classifier at Step II using best attributes

The following classifier (rules and classification performance measures) was obtained by ICRM2 algorithm starting from the data available at Stage II using the best attributes selected by the feature selection algorithms:

Rules for class 'Dropout':
1: IF (GPA < 8 AND Group IS NOT {C,K,O} AND Mother Studies IS NOT postgraduate
AND Job Time > 4 h) THEN 'Dropout' (Se 0.737 Sp 0.790 Cv 0.117)
2: IF (Alcohol IS {often,usually} AND Smoking IS yes) THEN 'Dropout' (Se 0.632 Sp 0.796 Cv 0.221)
3: IF (Size of the Classroom IS Large) THEN 'Dropout' (Se 0.509 Sp 0.986 Cv 0.368)

Rules for class 'Continue':
1: IF (AGE IS NOT HigherThan15 AND Alcohol IS {never,rarely}
AND Group IS NOT {A2,S,B2}) THEN 'Continue' (Se 0.710 Sp 0.842 Cv 0.632)
2: IF (GPA > 7.9 AND Mother Studies HIGHER Elementary
AND Size of the Classroom IS Small) THEN 'Continue' (Se 0.652 Sp 0.912 Cv 0.881)
3: IF (Job Time < 4 h AND Smoking IS No) THEN 'Continue' (Se 0.809 Sp 0.754 Cv 0.902)

Classification performance measures:
Confusion Matrix:
Actual vs. Predicted 'Continue' 'Dropout'
'Continue' 335 27
'Dropout' 10 47
Accuracy: 0.91
Geometric mean: 0.87
Correct predictions per class
Class 'Continue': 0.92
Class 'Dropout': 0.82

We can see that six IF–THEN rules were obtained: three for the 'Dropout' class and the other three for the 'Continue' class. For each rule, the sensitivity (Se), specificity (Sp) and coverage (Cv) are shown. Coverage measures the fraction of instances covered by the antecedent of a rule. As such, it is a measure of generality of a rule. It is also important to notice that a low coverage value (e.g. $0.1 < \text{coverage} < 0.3$) is normal in the context of unbalanced datasets because dropout students are the minority class. If we analyse the rules, we can see that there is a clear relation between student habits and social conditions and their final school performance/status. More specifically, these are as follows: a GPA of less than eight, a low level of the mother's education (mother studies), a time devoted to a job (job time) of more than 4 h per day, regular consumption of alcohol (alcohol), smoking habits (smoking) and large class groups of over 40 students (size of the classroom). These are indicators of a student's risk of dropout. On the other hand, a different attribute/indicator (not over 15 years old) has also appeared in the rules for detecting students that continue in the next semester of the course. Teachers can check all these conditions at Step II of the course for detecting students at risk of dropping out in time to help them. Regarding the classification performance measures, we can see that all the values obtained are high (greater than 80% in all the measures), and thus, it is a trustworthy model to use to classify new students in this early step of the course.

6.2. Classifier at Step V using all attributes

The next classifier (rules and classification performance measures) was obtained by ICRM2 algorithm at the end of the course using all the available attributes.

Rules for class 'Dropout'
1: IF (Maths IS 'F' and Computer Science IS BELOW 'B' (Se 0.930 Sp 0.992 Cv 0.341)
AND English IS BELOW 'A') THEN 'Dropout'
2: IF (Social Sciences IS BELOW 'D' AND Physics IS BELOW 'C') THEN 'Dropout' (Se 0.930 Sp 0.989 Cv 0.389)

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3: IF (Reading&Writing IS BELOW 'D') THEN 'Dropout' (Se 0.930 Sp 0.986 Cv 0.221)
4: IF (Humanities IS BELOW 'D' AND Physics IS 'F') THEN 'Dropout' (Se 0.930 Sp 0.981 Cv 0.157)

Rules for class 'Continue'
1: IF (Alcohol IS {never,rarely} AND Social Sci. IS NOT 'F' and Humanities IS NOT 'F') THEN 'Continue' (Se 0.942 Sp 0.982 Cv 0.812)
2: IF (Absenteeism IS 'NO' AND Maths IS NOT 'F' and Computer Science IS NOT 'F') THEN 'Continue' (Se 0.939 Sp 0.982 Cv 0.782)
3: IF (Reading&Writing IS NOT 'F' AND Expectative IS 'Will pass') THEN 'Continue' (Se 0.931 Sp 0.965 Cv 0.756)
4: IF (English IS NOT 'F' AND Physics IS NOT 'F') Then 'Continue' (Se 0.912 Sp 0.965 Cv 0.637)

Classification performance measures
Classification Confusion Matrix
Actual vs; Predicted 'Continue' 'Dropout'
'Continue' 362 0
'Dropout' 1 56
Accuracy: 0.99
Geometric mean: 0.98
Correct predictions per class
Class 'Continue': 1.00
Class 'Dropout': 0.98

We can see that eight IF–THEN rules were obtained, four for 'Dropout' class and the other four for 'Continue' class. If we analyse the rules about dropout, we can see now that the low scores achieved by students in each of the subjects of the course (Maths, Computer Science, English, Social Sciences, Physics, Reading&Writing and Humanities) are the only indicators of the dropout of the students. Nevertheless, it is interesting to see more indicators for detecting students who continue. For example, to abstain or only rarely consume alcohol (alcohol), not to skip class (absenteeism) and to have a high level of motivation or expectative (expectative) are indicators of a student who will continue in the next semester. About the classification performance measures, we can see that all the values obtained are near or equal to the highest possible value (100%). However, this precise classification model is not useful for making early prediction as it uses information gathered at the end of the course, when there is no time for any intervention regarding the students at risk of dropping out.

7. Related work and discussion

This work is related to two other fields engaged in the early predicting of student dropout and imbalanced data set. There are several works that apply data mining techniques for predicting dropout not only at the end of the course (described in Section 2) but also at early stages. For example, an EWS was developed using learning management system tracking data of a higher education course (Macfadyen & Dawson, 2010). They identified 15 variables demonstrating a significant simple correlation with final grade students at the University of British Columbia in 2008. Regression modelling generated a best-fit predictive model for this course, and a binary logistic regression analysis demonstrated the predictive power of this model (73.7% accuracy in week 7 and 81% in week 14, marking the terminus of the course). In other related work, a decision support system was developed for predicting success, excellence and retention from the student's early academic performance in a first year tertiary education programme (Mellalieu, 2011). This decision support system was based on several rules and regression equations derived from a test data set of student results from a previous delivery of the course. The results obtained were 69.6% accuracy in week 6 and 80.5% in week 12 (end of the course). In another work, several classification methods provided by WEKA (ZeroR, NB, SMO, IB1, OneR, PART and J48) were used for early prediction of student dropout in the Masaryk University (Bayer *et al.*, (2012)). They enriched the student data with information about the students' social behaviour gathered from email and discussion board conversations. They used sociograms and social network analysis to obtain new information about the student from the network such as neighbours' characteristics. They concluded that four semesters is the period at which their model can predict a dropout with high probability (83.22% using SMO) versus the final prediction after the seventh and last semester (91.11% using PART). Finally, if we compare these approaches with our proposal, the highest accuracy was obtained by our proposed ICRM2 algorithm both at the end of the course (99.8% in week 14) and even before the middle of the course (85.7% accuracy in week 4).

There are previous studies on the use of evolutionary computation and GP that address imbalanced data support and increase the motivation and justification of our proposal. Evolutionary-based algorithms have shown the good performance and adaptation of these algorithms to

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handle imbalanced data appropriately (Orriols-Puig & Bernadó-Mansilla, 2009). Specifically, GP also proved to be an efficient approach to resolve imbalanced data issues (López *et al*., 2013b). Defining a fitness function capable of dealing with imbalanced data is essential for achieving good performance on such data, paying special attention to the balance and trade-off between sensitivity and specificity for imbalanced classes (Patterson and Zhang, 2007).

However, there is scarcely any work regarding classification with imbalanced educational data, with the exception of our two previous papers. In a first paper (Márquez-Vera *et al*., 2011), we studied which indicators are most related to dropout in middle education using the most traditional classification algorithms. We used a data set with 670 students of whom 60 drop out, and we obtained the best performance when using JRip algorithm (87.5% GM and 96% accuracy). In a second paper (Marquez-Vera *et al*., 2013), we also used the same data set but proposed the application of different data mining approaches to deal with high dimensional and imbalanced data. We obtained the best performance when using cost-sensitive classification with JRip algorithm (94.6% GM and 96% accuracy). In this paper, we explored the specific problem of early dropout prediction in order to develop an EWS. Hence, one important difference as compared with our previous work is that this time, we only used the data gathered at each step of the course and at the specific moment/date in which they are obtained. Thus, we need to use a different data set that provides information about the student in each step of the course, in this case 419 students of whom 57 drop out. If we compare the results of our two previous approaches versus our current proposal, the highest accuracy was obtained when using our ICRM2 algorithm (99.1% GM and 99.8% accuracy).

There are some other interesting issues about the paper for discussion:

- It has been possible to reduce the number of attributes used in each step by selecting the best attributes for predicting dropout. And this fact is very important regarding our problem, because it allows us to save time and to reduce the amount of information needed to be collected. For the purposes of this study, all the information about students was collected for the sole purpose of this research from different sources (administration, teachers, parents, etc.) and different formats (papers, database, text files, excel files, etc.). And it is an arduous and time-consuming task to gather, integrate, pre-process and transform all this information into a suitable format ready to be used by a data mining algorithm. However, we obtained very high prediction of dropout using only a subset of attributes in all steps of the course. For example, the model discovered by ICRM2 algorithm at Step II when using the selected attributes (only 10 attributes) obtained an accurate enough value for making an early prediction of student dropout, very similar to the model obtained at Step III when using all attributes (27 attributes). These 10 attributes were as follows: GPA in secondary school, classroom/group enrolled, number of students in the group/class, age, attendance during morning/evening sessions, having a job, mother's level of education, time in 1000-m race, regular consumption of alcohol and smoking habits. It is also important to note that the factors that can affect low student performance may vary greatly depending on the student's educational level. In other words, certain factors that are vital in compulsory education might not be so important in higher education and vice versa. Thus, in order that we may adapt our methodology to suit a different domain, it is therefore firstly necessary to widely research all the possible factors.
- The execution time of the GP algorithm is not as high as could be expected. The proposed ICRM2 algorithm obtained the best results for predicting dropout in all the cases and steps of the course within a reasonable time frame. The execution times reported in Tables 5 and 7 show that all algorithms run fast, but ICRM2 is known to perform slower because of its genetic-based nature. However, in spite of its evolutionary learning process, the runtime of ICRM2 is lower than 30 s, which is acceptable for end users. Therefore, it is not a significant disadvantage. Were we to speed up the algorithm to reduce computing time further, we could make use of parallelization strategies using graphics processing unit computing, which are commonly employed in data mining and machine learning (Cano *et al*., 2012). Because evaluating the rules is the most time-consuming task in evolutionary rule learning, graphics processing units have sped up the process of accelerating rules evaluation in parallel.
- The low number of instances in the minority class is not a problem to prove the effectiveness of the classification algorithms. University of California, Irvine machine learning repository imbalanced data sets commonly used in literature (Fernández *et al*., 2008) are categorized with regard to the IR: low imbalance for IR lower than 3, medium imbalance for IR between 3 and 9 and high imbalance for IR higher than 9. The data set that we employ has 57 dropouts from 419 students. Its IR of 6.35 is such that it is considered as a data set with medium imbalance. On the other hand, in our two previous related works, the data sets employed have a similar number of students who drop out (60 students). However, its IR of 10.16 is such that it is considered as a data set with high imbalance. In the future, we hope to carry out more experiments using a greater number of educational data sets in order to test the results obtained with our ICRM2 algorithms when using different numbers of dropout students and different IR.

8. Conclusions

The proposed methodology has shown to be valid for predicting high school dropout. We carried out two experiments using data from 419 first year high school Mexican students. It is important to notice that most of the current research on the application of EDM to resolve the problems of student dropouts has been applied primarily to the specific case of higher education. However, little research into compulsory education dropout rates has been conducted, and what has been

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found uses only statistical methods, not data mining techniques. This work discovers classification models trustworthy enough to make an early prediction of dropout, before the middle of the course. In fact, we obtained good results of predictions in Steps II and III, that is, at the first 4 and 6 weeks of the course, respectively. Our proposed ICRM2 algorithm outperformed all the traditional classification algorithms used, not only in TN rate but also in GM that measures in a balanced way the accuracy of predicting dropout (TN rate) and continue (TP rate). In addition, ICRM2 algorithm provides a white-box model that is very comprehensible for a human user. The discovered IF–THEN rules show the indicators and relationships that cause a student to continue with school or drop out, and they can therefore be used in the decision-making process as what happens in EWS. Therefore, the obtained models can be used to detect students at risk of dropping out as soon as possible, and stakeholders can provide the appropriate advice to each student before the end of the course. In this line, it is important to realize that identifying students at risk of dropping out by using an EWS is only the first step in truly addressing the issue of school dropout. The next step is to identify the specific needs and problems of each individual student who is in danger of dropping out and then to implement programmes to provide effective and appropriate dropout-prevention strategies. Therefore, stakeholders should be able to attend to students' needs to help them in time to avoid dropout. For example, some possible responses to early warning signals are as follows: informing and involving parents, creating multi-disciplinary support teams and individual action plans, fines/sanctions/prosecution, etc. So, in the future, we want to develop this intervention part of the dropout EWS in high school. We would also like to be able to evaluate the effect of these different types of interventions to find which are the most appropriate for each type of students at risk of dropout. However, in order to do so, it is necessary to gather information about the results obtained after applying these processes over several classes of students.

Appendix: Pseudocode of the ICRM2 algorithm

ICMR2 algorithm

BEGIN
1. To initialise classifier
   It creates two empty rule-set: one for dropout class and another for success class.
2. To obtain the rules for predicting the dropout class
   It runs GBGP algorithm in order to obtain a set of rules that predicts the success class
3. To obtain the rules for predicting the success class
   It runs GBGP algorithm in order to obtain a set of rules that predicts the success class
4. To obtain the final rule-base classifier
   It combines the rule-sets for the dropout and the success classes to build the final classifier

END

GBGP algorithm

BEGIN
WHILE there are remaining instances and the number of rules is lower than the maximum allowed REPEAT
1.1. To initialise the population for a class
   It generates the individuals (rules) of the population to learn a given class.
1.2. To do parent selection
   It selects the parents on which the genetic operations are applied
1.3. To do crossover
   It mixes the genetic information of the parents to generate an offspring
1.4. To do mutation
   It mutates the offspring to facilitate the search exploration
1.5. To evaluate
   It evaluates the fitness of the new rules
1.6. To update population
   It selects the best rules from the parent population and the offspring to keep the population size constant with the best rules for the next generation
1.7. IF the algorithm has more generations to run
   GOTO step 1.2 and iterate next generation
ELSE
   CONTINUE with step 1.8
1.8. To select the best rule and append to the rule-set
   It selects the best rule from the population using the fitness function and appends to the rule-set
1.9. To remove the instances covered by the rule
   Instances covered by the rule are removed from the training examples so that new rules can be learned on the remaining instances
END WHILE
2. To return the rule-set
END

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